THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Chemical Reactions and Equations** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Formation of new substance indicates chemical change.
2	**(d)** 44 g	Mass is conserved: 12 + 32 = 44 g.
3	**(b)** copper	CuO is reduced to Cu by hydrogen.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Energy is needed to break bonds.
5	**(a)** 16 g	40 - 24 = 16 g.
6	**(d)** zinc	Zn is more reactive; $ZnSO_4$ is colourless.
7	**(c)** A is true but R is false.	A is true; R is false because nitrogen is unreactive.
8	**(b)** 36 g	4 + 32 = 36 g.
9	**(d)** Na_2CO_3	Carbonates release CO_2.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Equal atoms mean equal mass.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Mass of CO_2 = 12 + 32 = 44 g (1); Law of conservation of mass (1).	1+1
12	Gases NO_2 and O_2 escape (1); Only solid PbO remains; the total mass is still conserved (1).	1+1
13	$ZnCO_3 \longrightarrow ZnO + CO_2$ (1); Carbon dioxide (turns lime water milky) (1).	1+1
14	Iron (56) dissolves and copper (63.5) is deposited (1); Copper atoms are heavier, so the net mass rises (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Equation already balanced: $PbO + C \longrightarrow Pb + CO$ (1); PbO is the oxidising agent (1); C is the reducing agent (1).	1+1+1
16	Basic copper carbonate (1); Copper reacts with moisture, CO_2 and oxygen (1); Prevent by lacquer/paint or keeping dry (1).	1+1+1
17	Hydrogen gas (1); Burns with a pop sound near a flame (1); $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (1).	1+1+1
	OR: $Fe_3O_4 + 4\,H_2 \longrightarrow 3\,Fe + 4\,H_2O$ (2); H_2 is oxidised (1).	2+1
18	X is magnesium, Y is magnesium oxide (1); $2\,Mg + O_2 \longrightarrow 2\,MgO$ (1); Y is basic as $Mg(OH)_2$ forms with water (1).	1+1+1
	OR: Brown copper turns black: $2\,Cu + O_2 \longrightarrow 2\,CuO$ (1); Black CuO turns brown again (1); $CuO + H_2 \longrightarrow Cu + H_2O$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	A: no rust (1); B: no rust (1); C: rust forms (1); Conclusion: air (oxygen) and water are both needed (1); Prevention: galvanisation/painting (1).	1+1+1+1+1
	OR: Green to white (water lost), then reddish brown residue (1); $2\,FeSO_4 \longrightarrow Fe_2O_3 + SO_2 + SO_3$ (1); Gases: SO_2 and SO_3 (1); Smell of burning sulphur (1); Thermal decomposition (1).	1+1+1+1+1
20	Zinc: bubbles of hydrogen (1); $Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$ (1); Copper: no reaction (1); Zinc is above hydrogen in the reactivity series (1); Copper is below hydrogen so cannot displace it (1).	1+1+1+1+1
	OR: (i) Zn is more reactive (1); (ii) Copper is deposited on the nail (1); (iii) Silver is formed by photolysis (1); (iv) Basic copper carbonate forms in moist air (1); $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$; $2\,AgCl \longrightarrow 2\,Ag + Cl_2$ (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Chemical Reactions and Equations** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 40 mL	Ratio H2 : O2 = 2 : 1.
2	**(b)** increases slightly	Heavier Cu (63.5) replaces Fe (56) on the nail.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Reactivity decides displacement.
4	**(d)** 23	Al2(SO4)3: 2+3+12 = 17; 3H2: 6; total 23.
5	**(d)** hydrogen	Hydrogen explodes with a pop.
6	**(a)** Both A and R are true and R is the correct explanation of A.	Photolysis gives grey silver.
7	**(d)** 12	$3\,Ca(OH)_2 + 2\,H_3PO_4 \longrightarrow Ca_3(PO_4)_2 + 6\,H_2O$: 3+2+1+6=12.
8	**(c)** tube (iii) only	Needs both air and water.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Oxidation of glucose releases energy.
10	**(c)** 44 g	100 - 56 = 44 g.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	H_2S loses hydrogen: oxidised (1); Cl_2 gains hydrogen: reduced (1).	1+1
12	$CaO + H_2O \longrightarrow Ca(OH)_2$ (1); $Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$ (1).	1+1
13	Correct formulae (1); $Na_2SO_4(aq) + BaCl_2(aq) \longrightarrow BaSO_4(s) + 2\,NaCl(aq)$ (1).	1+1
14	Silver reacts with hydrogen sulphide in air (1); Silver sulphide (Ag_2S) (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$2\,Mg + SO_2 \longrightarrow 2\,MgO + S$ (2); SO_2 is reduced (loses oxygen) (1).	2+1
16	X is quick lime, CaO; $CaCO_3 \longrightarrow CaO + CO_2$ (1); $CaO + H_2O \longrightarrow Ca(OH)_2$ (1); Y is slaked lime, an exothermic product used for whitewash (1).	1+1+1
17	White precipitate of AgCl (1); $AgNO_3 + NaCl \longrightarrow AgCl + NaNO_3$ (1); Double displacement (precipitation) (1).	1+1+1
	OR: $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$; 65 g Zn gives 2 g H2 (1); 6.5 g Zn = 0.1 mol (1); Mass of H2 = 0.2 g (1).	1+1+1
18	Low temperature slows oxidation (rancidity) (1); Antioxidants (1); Airtight packing / nitrogen flushing (1).	1+1+1
	OR: $2\,Mg + O_2 \longrightarrow 2\,MgO$: two elements give one compound (1); Mg gains oxygen: oxidised (1); O_2 is reduced (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$2\,H_2O \longrightarrow 2\,H_2 + O_2$: 36 g gives 4 g H2 and 32 g O2 (1); 9 g gives 1 g hydrogen (1); and 8 g oxygen (1); Total 9 g (conservation of mass) (1); Volume ratio H2 : O2 = 2 : 1 (1).	1+1+1+1+1
	OR: (a) Exothermic: heat given out (1); (b) Endothermic: sunlight absorbed (1); (c) Endothermic: heat absorbed (1); (d) Exothermic: energy released (1); (e) Exothermic: beaker becomes hot (1).	1+1+1+1+1
20	$CaCO_3 \longrightarrow CaO + CO_2$ (1); CaO = 56 kg (1); CO2 = 44 kg (1); Thermal decomposition (1); Quick lime used in making cement/whitewash (1).	1+1+1+1+1
	OR: Flask weighed with both solutions kept apart (1); Mix them by tilting (1); Yellow precipitate of PbI_2 forms (1); Total mass is unchanged (1); Law of conservation of mass is verified (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Chemical Reactions and Equations** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** Cu^{2+}	Cu2+ gains electrons/ is converted to Cu.
2	**(c)** A is true but R is false.	A true; R false because there is no oxidation or reduction.
3	**(d)** 3 g	Total mass 15 g; 15 - 12 = 3 g.
4	**(b)** magnesium or zinc	More reactive metals corrode in preference.
5	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true, but the ratio is explained by formula H_2O, not merely being a compound.
6	**(d)** 1.1 g	50.0 - 48.9 = 1.1 g.
7	**(d)** changed the formula of the product, which is not allowed	Balancing is done by coefficients only, not by altering formulae.
8	**(d)** A is false but R is true.	Mg is oxidised, so A is false; R is true.
9	**(c)** 20 mL	3 volumes H2 : 2 volumes NH3.
10	**(b)** $CaCO_3$	$CaCO_3 \longrightarrow CaO + CO_2$; $CaO + H_2O \longrightarrow Ca(OH)_2$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Exothermic reaction (heat is released) (1); e.g. $CaO + H_2O \longrightarrow Ca(OH)_2$ (1).	1+1
12	$2\,Mg + CO_2 \longrightarrow 2\,MgO + C$ (1); Mg is oxidised (CO_2 is reduced) (1).	1+1
13	Iron is more reactive and displaces copper (1); $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$ (1).	1+1
14	Volume of hydrogen = 20 mL (1); $2\,H_2O \longrightarrow 2\,H_2 + O_2$ gives 2 : 1 by volume (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	100 g gives 56 g CaO and 44 g CO2 (1); CaO = 5.6 g (1); CO2 = 4.4 g (1).	1+1+1
16	Zinc is more reactive than copper (1); $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$ (1); A less reactive metal cannot displace a more reactive one (1).	1+1+1
17	$CaO + H_2O \longrightarrow Ca(OH)_2$ releases heat: exothermic (1); $CaCO_3 \longrightarrow CaO + CO_2$ absorbs heat: endothermic (1); Two reactions are opposite in direction/energy (1).	1+1+1
	OR: Molar mass AgCl = 143.5 g; 2AgCl -> 2Ag + Cl2 (1); 14.35 g = 0.1 mol (1); Silver = 0.1 x 108 = 10.8 g (1).	1+1+1
18	Subscripts must not be changed in balancing (1); H_2O_2 is a different compound (hydrogen peroxide) (1); Correct: $2\,H_2 + O_2 \longrightarrow 2\,H_2O$ (1).	1+1+1
	OR: Oxygen oxidises iron (1); Moisture speeds up rusting / provides electrolyte (1); Zinc forms a protective layer and corrodes first (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Mg oxidised, O_2 reduced (1); (i) Mg is the reducing agent (1); (ii) H_2S oxidised, Cl_2 reduced (1); (ii) H_2S is the reducing agent (1); (iii) C oxidised, ZnO reduced; C is the reducing agent (1).	1+1+1+1+1
	OR: Painting/oiling keep out moisture and air (1); Galvanisation uses zinc coating (1); Alloying (stainless steel) (1); Zinc is more reactive and corrodes first (1); So iron is protected even if the surface is damaged (1).	1+1+1+1+1
20	(a) $4\,Fe + 3\,O_2 \longrightarrow 2\,Fe_2O_3$, combination (2); (b) $2\,NH_3 + 3\,CuO \longrightarrow 3\,Cu + N_2 + 3\,H_2O$, redox (1); (c) $8\,Al + 3\,Fe_3O_4 \longrightarrow 4\,Al_2O_3 + 9\,Fe$, displacement/redox (2).	2+1+2
	OR: Thermal: $CaCO_3 \longrightarrow CaO + CO_2$ (heat) (1); Electrolytic: $2\,H_2O \longrightarrow 2\,H_2 + O_2$ (electricity) (1); Photolytic: $2\,AgBr \longrightarrow 2\,Ag + Br_2$ (light) (1); Application of photolytic: photography (1); Application of electrolytic: extraction of hydrogen/oxygen (1).	1+1+1+1+1

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